NORTH SOUTH UNIVERSITY |
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING |

Major: Computer Science and Engineering
Course Name: Programming Language I
Course Code: CSE115
Course Instructor Initial: MSRb
Section: 5

GROUP 1: THEORY TERM PROJECT FINAL REPORT

Project Name: **Text-Based Adventure Game**

GOAL: To create an interactive text-based adventure game where players navigate through a story by making choices and solving puzzles.

GROUP MEMBERS:

	Group Member Name	**Student ID**
1.	Chowdhury Rounak Jahan Bushra	2612545042
2.	Md. Shaharear Hossain Naffiz	2611327042
3.	Smita Elizabeth Biswas	2611961042
4.	Sumaiya Karim Ilma	2532746042

ABSTRACT

The project presents a text-based interactive adventure game that simulates a student's first day at North South University. The main idea is to create an engaging system where players navigate through the most interesting locations in the university, make decisions according to their situation and solve puzzles along the way. The game is developed using C programming language focusing mainly on loops, conditional statements, and functions to allow a smooth flow in the story. Different choices lead to different outcomes making the experience dynamic and personal.

INTRODUCTION

The transition from school life to university life can be both exciting and challenging. This project is based on imitating a student's first day at NSU through an interactive text-based adventure game. The purpose of this project is to demonstrate how programming concepts learnt in CSE115 course can be effectively used to create a decision based storytelling system.

This game allows users to make choices such as attending classes, interacting with seniors, classmates, and faculties, and exploring the extensive university campus. As this game gives a glimpse of life at NSU, this could possibly help motivate students to consider NSU for further studies.

FEATURES

Feature 1: Interactive Storyline

This game allows users to navigate through the following scenes:

1. North South University Main Gate
2. Academic Building
3. Central Library
4. Club Fair
5. NSU Plaza Area

Each of the aforementioned scenes presents choices that influence game progression and affect the outcome.

Feature 2: Player Statistics System

Three stats are maintained and printed at the end of the game:

1. Knowledge
2. Confidence
3. Mistakes

They are kept in the integer variables knowledge, confidence, and mistakes respectively.

Feature 3: The Multiple Endings

The final result depends on the accumulated stats.
if (knowledge >= 6 && confidence >=6)

Feature 4: Input Validation

This ensures only valid inputs are accepted.
if (scanf("%d",&choice))==1 && choice>=min && choice<=max)

Feature 5: The Modular Design

Each scene is implemented as a separate function, which are as follows:

1. gate()
2. academicBlock()
3. libraryPuzzle()
4. clubFair()
5. plaza()

CONCEPTS USED

The project is an implementation of several fundamental programming concepts from our CSE115 course.

1. Variables and Data Types: int, char
2. Conditional Statements: if-else and switch
3. Loops: Conditional Loops (while)
4. Arrays and Strings: to store and manipulate data such as player name
5. Global Variables: the player stats

INDIVIDUAL MEMBER CONTRIBUTION

1. Chowdhury Rounak Jahan Bushra
Student ID: 2612545042

Rounak worked on the overall program design and flow. She was involved in breaking down the code into several functions and making sure the game is executed properly. She determined the structure of functions and how the parts of the game interact with each other.

She wrote the main function and created the main menu, enabling users to navigate through the main menu to either start a game, read the instructions and exit the game. Additionally,

she managed the game start-up process, such as resetting the stats of the player and getting the name of the player before starting the game.

She handled scene transition control, linking the different scenes in the game in a coherent order (Gate → Academic Building → Library → Club Fair → Plaza → Ending) to provide a seamless narrative. Also, she added input validation for user inputs, to avoid any unexpected errors while running the game. In essence, her work ensured the program was well-organised, easy to use and functional.

2. Md. Shaharear Hossain Naffiz
Student ID: 2611327042

Naffiz was responsible for the technical implementation of the input system and managing user interaction within the game. His contribution focused on how the program processes and responds to user inputs, ensuring smooth execution during gameplay.

A key aspect of his work was developing the input validation mechanism through the getValidChoice() function which ensures only valid numeric values within specified range are accepted as inputs. It uses a while loop to repeatedly prompt the user until correct input is entered, and clears the input buffer using getchar() when invalid data is detected. This prevents unexpected behaviour and improves stability.

Additionally, he handled user input collection using scanf(), applying safe format specifiers such as "%49s" to avoid buffer overflow when storing strings like player names. He also worked on integrating user inputs with control logic using switch-case statements, ensuring that user choices were correctly executed during runtime. He contributed to testing and debugging, focusing on input related errors and edge cases. Overall, his work strengthened the program's readability, accuracy, and user interaction experience.

3. Smita Elizabeth Biswas
Student ID: 2611961042

Smita was responsible for creating a draft of the storyline that the game follows. She implemented parts of Scene 2 (Academic Building) and Scene 3 (Central Library) of the game.

In Scene 2, she developed the decision-making logic where the player makes their choice among the different options, and outcomes are determined using conditional statements. An if-else structure was used to handle two main options: the lift or stairs. Choosing the lift increases the confidence variable. If the player chooses stairs, an additional nested if-else is used to evaluate the player's answer to a question asked by the professor. Based on the accuracy of the answer, the program either updates the knowledge or mistakes variable. This scene includes input validation which was handled using the getValidChoice() function,

which uses a while loop to repeatedly prompt the user until a valid input within a specific range is entered.

In Scene 3, she implemented a puzzle that requires the player to calculate the total number of credits. The program collects input using scanf(), and uses an if-else condition to verify the answer. If correct, the knowledge variable is incremented, otherwise the mistakes variable is increased. She also ensured smooth progression between scenes by using structured function calls such as libraryPuzzle() and clubFair(), maintaining continuity in gameplay.

4. Sumaiya Karim Ilma
Student ID: 2532746042

Ilma was responsible for testing, debugging, and refining the overall gameplay experience. Her contribution focused on ensuring that the program runs smoothly and logically under different conditions.

She conducted systematic testing of the game by running multiple scenarios with different user choices to identify logical errors and unexpected behaviours, giving special attention to verifying whether the three variables updated correctly under different choices. She checked whether scene transitions occurred as intended and made sure no part of the game flow was broken during execution.

Additionally, she worked on output clarity which included adjusting prompts, feedback messages, and instructions to make them more understandable. She also assisted in debugging issues related to incorrect outputs, misplaced conditions, and minor logical errors.

As a whole, her contribution ensured that the final program was polished, user-friendly, and functionally consistent.

TESTING AND RESULTS

Test Case 1:

Input: Invalid input (letters/ symbols instead of numeric value)
Expected Output: Error message and retry
Result: Successful

```
  1. Start New Game
  2. How to Play
  3. About This Project
  4. Exit

Your choice: A
[!] Invalid input. Please enter a number between 1 and 4: 7
[!] Invalid input. Please enter a number between 1 and 4: 1
```

Test Case 2:

Input: Valid input (numeric value within range)
Expected Output: Smooth Progression
Result: Successful

```
============================================================
                NSU CAMPUS ADVENTURE
            A Text-Based Adventure Game
                CSE115 - Group Project
============================================================

  1. Start New Game
  2. How to Play
  3. About This Project
  4. Exit

Your choice: 1


Enter your name: SMITA

============================================================
 Welcome to North South University, SMITA!
============================================================

Today is your very first day on campus.
Every choice you make will shape your experience.
Good luck!
```

```
============================================================
                SCENE 1 : NSU MAIN GATE
============================================================

You arrive at the main entrance of North South University.
A security guard stops you at the gate.

Guard: "This is a restricted campus. Where is your
        admission proof?"

What do you do?
  1. Be honest and explain you are a new student.
  2. Confidently state that you belong here.
  3. Show your admission confirmation email.

Your choice: 3

You show your email on your phone. The guard
verifies it and compliments your preparedness.
[+1 Knowledge, +1 Confidence]
```

```
============================================================
            SCENE 2 : SAC ACADEMIC BUILDING
============================================================

You have located the SAC building where your CSE115
class is held. You need to reach the 4th floor.

How do you get there?
  1. Take the lift.
  2. Take the stairs.

Your choice: 2

On the stairwell you bump into your professor!
She smiles and asks you a quick warm-up question.

Professor: "Where do programmers primarily study
            and practice their code at NSU?"

  1. The Computer Lab
  2. The Central Library
  3. The Cafeteria

Your answer: 2

Professor: "Excellent! The library has dedicated
programming resources. I will see you in class."
[+2 Knowledge]
```

```
============================================================
                SCENE 3 : CENTRAL LIBRARY
============================================================

After class you head to the Central Library to print
your course schedule. The self-service kiosk asks you
to solve a quick verification puzzle.

Puzzle: CSE115 carries 3 credits. If you are enrolled
        in 4 courses this semester, what is your total
        credit load?  (3 x 4 = ?)

Your answer: 12

Correct! The kiosk unlocks and prints your schedule.
[+2 Knowledge]
```

```
============================================================
                SCENE 4 : CLUB FAIR
============================================================

Outside the library you discover the annual NSU Club
Fair in full swing. Dozens of stalls line the walkway.

Which club catches your interest?
  1. NSU Programming Club - competitive coding team.
  2. NSU Cultural Club    - events, arts & heritage.
  3. Skip the fair and walk toward the Plaza.

Your choice: 1

You sign up for the Programming Club. Senior
members immediately share study resources.
[+2 Knowledge, +1 Confidence]
```

```
============================================================
                SCENE 5 : NSU PLAZA
============================================================

The sun is lower in the sky as you reach the iconic
NSU fountain plaza. Students are chatting and relaxing
after their classes. It is the perfect end to a long day.

What do you do for the rest of the afternoon?
  1. Join a group of students chatting by the fountain.
  2. Visit the Registrar's Office to learn about
     add/drop, grading policy, and advising hours.
  3. Head home - it has been a big enough day.

Your choice: 2

The Registrar's staff walk you through key academic
policies. You leave feeling informed and prepared.
[+3 Knowledge]

============================================================
              END OF YOUR FIRST DAY AT NSU
============================================================
```

```
FINAL STATS
------------
Knowledge  : 7
Confidence : 5
Mistakes   : 0

RESULT
------
*** PROMISING STUDENT ***
You had a solid first day. Keep up the momentum
and NSU will treat you very well.

Thank you for playing, Smita!
Good luck with your semester at North South University.
```

Test Case 3:

Input: Different choice combinations
Expected Output: Different ending
Result: Successful

```
FINAL STATS
------------
Knowledge  : 5
Confidence : 1
Mistakes   : 1

RESULT
------
*** PROMISING STUDENT ***
You had a solid first day. Keep up the momentum
and NSU will treat you very well.

Thank you for playing, Bushra!
Good luck with your semester at North South University.
```

Same scenes as shown in the screenshots above except the output is different with different confidence, knowledge, and mistakes variables.

The program runs perfectly and produces different outcomes based on user choices. The video that has been submitted also uses the same test values.

CHALLENGES AND SOLUTIONS

1.Input Handling
Problem : Invalid user input caused issues and unexpected outputs.
Solution: Implemented validation function getValidChoice() and getChoice().

2.Managing Game Flow
Problem: Keeping track of scenes.
Solution: Used function calls to control flow.

3.Designing Multiple Endings
Problem: Balancing conditions.
Solution: Used conditional statements based on stats.

CONCLUSION

This project successfully demonstrates how a text-based game can be created using basic C programming concepts learnt in the CSE115 course. It improves understanding of functions, control structures, and user interaction.

What we learned:

- Function-based program design
- Input validation
- Decision-making logic
- Implementation of loops and conditional statements

Future Improvements:

Concepts like pointers and file handling could be used, but as we learned these concepts slightly later in the semester, we could not implement them in our already modeled project. In the future projects we can implement these concepts making the game more dynamic.

Adding more scenes and story branches would make the story more adventurous and engaging. Also, having a saving/loading feature would give more clarity to the player. Having a scoring system or levels would make the game even more exciting.

Nevertheless, we tried our best to use the concepts we learned in the classes throughout the duration of our course. Next time we will definitely make use of the concepts we missed out this time, and make an advanced version of this game.